

Iago Nascimento Mamede

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Education

Institute of Physics, University of São Paulo <i>Bachelor's Degree in Physics</i>	January 2018 – December 2021 São Paulo, Brazil
Institute of Physics, University of São Paulo <i>Master's Degree in Physics</i>	January 2022 – May 2024 São Paulo, Brazil
Institute of Physics, University of São Paulo <i>PhD in Physics</i>	June 2024 – Present São Paulo, Brazil

Professional Background

Undergraduate Research <i>Institute of Physics, University of São Paulo</i>	October 2018 – August 2019 São Paulo, Brazil
• Competition between individual and collective dynamics in models of economic agents • Description: The project consisted of studying a model (Schelling) that draws a parallel with aspects of statistical physics for the study of segregation among individuals. The study aimed to analyze aspects of the collective and individual dynamics of this model. In general, we used methods for the analysis of phase transitions to understand the changes brought about by increasing the degree of cooperativity among individuals, leading to the maximization of the utility function. This function has a simple character and, through its introduction, prepares the ground for a more complex and mathematically deeper analysis. • Status: Completed; • Nature: Research; • Fellowship: CNPq; • Students involved: Undergraduate (1); • Members: Iago Nascimento Mamede - Member / André de Pinho Vieira - Coordinator;	
Undergraduate Research <i>Institute of Physics, University of São Paulo</i>	September 2019 – January 2020 São Paulo, Brazil
• Percolation through aperiodic layers • Description: The project involved the investigation of models constructed according to the construction of two-dimensional aperiodic networks. This investigation made use of two main tools: computational simulations and an approximate real-space renormalization-group treatment. The goal was to evaluate how different substitution sequences, inducing geometrical fluctuations of distinct nature, affect the universality class to which the percolation transition in these networks belongs. • Status: Deactivated; • Nature: Research; • Fellowship: CNPq; • Students involved: Undergraduate (1); • Members: Iago Nascimento Mamede - Member / André de Pinho Vieira - Coordinator;	
Undergraduate Research <i>Institute of Physics, University of São Paulo</i>	September 2020 – November 2021 São Paulo, Brazil
• Stochastic thermodynamics and efficiency in interacting periodic systems	

- **Description:** This undergraduate research project aimed to study the thermodynamic properties and efficiency of Brownian particles subjected to periodic forces. Such systems have attracted considerable interest because they can be used as starting points for the description of different problems in technological areas such as particle transport in nanotechnology and biotechnology, heat conduction in engineering, chemical reactions, biological systems, and others. Our study aimed to investigate the efficiency of finite-time heat engines composed of both noninteracting and interacting Brownian particles, taking into account the role of different ingredients such as oscillating forces, period, phase lag, interaction between particles, and other parameters. We investigated which protocol proves to be more efficient: noninteracting particles of negligible mass (overdamped Brownian particles), noninteracting particles (underdamped case), or interacting particles.
- **Status:** Completed;
- **Nature:** Research;
- **Fellowship:** FAPESP;
- **Students involved:** Undergraduate (1);
- **Members:** Iago Nascimento Mamede - Member / Carlos Eduardo Fiore dos Santos - Coordinator;
- **Results:** Results published in Phys. Rev. E 105, 024106 – Published 4 February 2022

BEPE Master's Fellowship

May 2023 – August 2023

Niels Bohr Institute

Copenhagen, Denmark

- **Thermodynamics of interacting heat engines and Landauer's principle**
- **Description:** The bilateral master's project (BEPE), aimed at advancing the field of nonequilibrium thermodynamics and strengthening the scientific collaboration between our research group and the Niels Bohr Institute of Physics, is divided into two parts. First, we intended to study the thermodynamic properties of machines composed of interacting quantum dots (QDs) beyond the mean-field case, which had remained unexplored, in order to verify how collective effects can improve the performance of heat engines. We addressed the role of network topology and how cooperative effects in such cases can enhance power and efficiency. Our analysis was carried out for homogeneous and heterogeneous structures, as well as for regular and random regular networks. Second, we understood/established how collective effects, for example in systems composed of interacting Brownian units and the aforementioned QDs, can be used to generate a protocol of minimum excess entropy production in connection with Landauer's principle.
- **Status:** Completed;
- **Nature:** Research;
- **Fellowship:** FAPESP;
- **Students involved:** Graduate (1);
- **Members:** Iago Nascimento Mamede - Member / Carlos Eduardo Fiore dos Santos - Coordinator / Karel Proesmans - Coordinator;
- **Results:** Results to be published in PRR, available on arXiv:2308.02255. Another work concerning first-passage times is under submission to Journal of Physics A.

Master's Degree

April 2022 – May 2024

Institute of Physics, University of São Paulo

São Paulo, Brazil

- **Thermodynamics and efficiency of interacting systems: Study in harmonic chains, collisional approach, and phase transitions**
- **Description:** This master's dissertation project aimed to study the thermodynamic properties, efficiency, and *bounds* for heat engines composed of interacting particles/entities. Initiated during the candidate's undergraduate research project in the context of interacting Brownian particles, we intended to advance in three different directions: interacting Brownian particles in the presence of periodic drivings that promote heat-to-work conversion, interacting systems exhibiting phase transitions, and a collisional approach for Brownian particles, in which contact with the reservoir and the applied force occur sequentially. Finally, we intended to extend, compare, and propose new *bounds* for efficiency, power, and dissipation with precision for the above class of systems. This part of the project was developed through collaboration with Prof. Karel Proesmans, the candidate's co-supervisor.

- **Status:** In progress;
- **Nature:** Research;
- **Fellowship:** FAPESP;
- **Students involved:** Graduate (1);
- **Members:** Iago Nascimento Mamede - Member / Carlos Eduardo Fiore dos Santos - Coordinator / Karel Proesmans - Coordinator;
- **Results:** Results to be published in PRR, available on arXiv:2308.02255. Another work concerning first-passage times is under submission to Journal of Physics A.

PhD

June 2024 – Present

Institute of Physics, University of São Paulo

São Paulo, Brazil

- **Stochastic thermodynamics of collective heat engines: Finite reservoirs and active particles**
- **Description:** The present doctoral thesis project aims to study systems exhibiting collective behavior and is structured in two parts. The first part consists of the study of the thermodynamics and performance of collective systems composed of interacting Brownian particles subjected to nonlinear forces and systems described by master equations coupled to finite reservoirs. In the latter case, we intend to study the statistics of power, efficiency, and dissipation, and this part of the project is related to the thematic project 2022/15453-0, entitled "Correlated quantum materials", coordinated by Eduardo Miranda (UNICAMP), in which I am a principal investigator. The second part of this project focuses on the thermodynamics and performance of different models for active matter under two approaches: the first continuous, described by a Fokker-Planck equation, and the second involving systems with discrete symmetry. In the latter case, we intend to propose an "active" version of the system composed of interacting heat engines investigated in the candidate's master's project.
- **Status:** In progress;
- **Nature:** Research;
- **Fellowship:** FAPESP;
- **Students involved:** Graduate (1);
- **Members:** Iago Nascimento Mamede - Member / Carlos Eduardo Fiore dos Santos - Coordinator / Karel Proesmans - Coordinator;

PhD

March 2024 – Present

U Hasselt

Hasselt, Belgium

- **Stochastic thermodynamics of collective heat engines: Finite reservoirs and active particles**
- **Description:** The present project aims to investigate different classes of systems within the framework of stochastic thermodynamics. One of our main objectives is to characterize the performance of different classes of heat engines. The different procedures used to construct machines at the nanoscale are most often applied to systems composed of a single unit, such as quantum dots or individual Brownian particles. More recently, there has been growing interest in the performance of systems based on interacting machines that exhibit collective behavior, some of which have been studied in previous research, as well as those based on active particles. The main objective of this proposal is to study collective behavior in these two classes of systems.
- **Status:** In progress;
- **Nature:** Research;
- **Students involved:** Graduate (1);
- **Members:** Iago Nascimento Mamede - Member / Bart Cleuren - Supervisor;

List of Most Relevant Publications

- **Obtaining efficient thermal engines from interacting Brownian particles under time-periodic drivings**
Iago N. Mamede, Pedro E. Harunari, Bruno A. N. Akasaki, Karel Proesmans, and C. E. Fiore Phys. Rev. E 105, 024106 – Published 4 February 2022

- **Obtaining efficient collisional engines via velocity-dependent drivings**
Iago N. Mamede, Angel L. L. Stable, and C. E. Fiore Phys. Rev. E 106, 064125 – Published 19 December 2022
- **Thermodynamics of interacting systems: the role of the topology and collective effects**
Iago N. Mamede, Karel Proesmans, Carlos E. Fiore Phys. Rev. Research 5, 043278 – Published 22 December 2023
- **Work statistics at first-passage times**
Iago N Mamede, Prashant Singh, Arnab Pal, Carlos E. Fiore, and Karel Proesmans New Journal of Physics, Volume 26, March 2024
- **Steady-state heat engines driven by finite reservoirs**
Iago N Mamede, Saulo V Moreira, Mark T Mitchison, Carlos E Fiore Physics Journal E 2026/01
- **Collective heat engines via different interactions: Minimal models, thermodynamics, and phase transitions**
Iago N Mamede, Vitória T Henkes, Carlos E Fiore Physics Review E

Links

- ORCID: <https://orcid.org/0000-0003-2380-4012>
- Google Scholar:
<https://scholar.google.com/citations?user=IzISrFEAAAAJ&hl=pt-BR&oi=sra>
- Website: <https://iagomamede.github.io>

Additional Information

- Jayme Tiomno Summer School (2018) - Institute of Physics, University of São Paulo;
- Jayme Tiomno Summer School (2019) - Institute of Physics, University of São Paulo;
- XIII CBPF School (2021) - Quantum Thermodynamics Course;
- XIII CBPF School (2021) - Topics in Open Quantum Systems Course;
- XIII CBPF School (2021) - Phase Transitions and Critical Phenomena Course;
- Awarded as "Outstanding Research by an Early-Career Student" at the 2nd Graduate Symposium of the Institute of Physics, University of São Paulo;
- Seminar presentations during my stay at the Niels Bohr Institute, Copenhagen, Denmark;
- Participation in the Workshop on Soft Matter, IFT-Unesp, São Paulo, Brazil;
- Presentation of work at the international conference **20th StatPhys, Florence, Italy**;
- Presentation of work at the international conference **Fluctuations in Self-Interacting and Learning Processes (satellite meeting of StatPhys29), Stockholm, Sweden**;